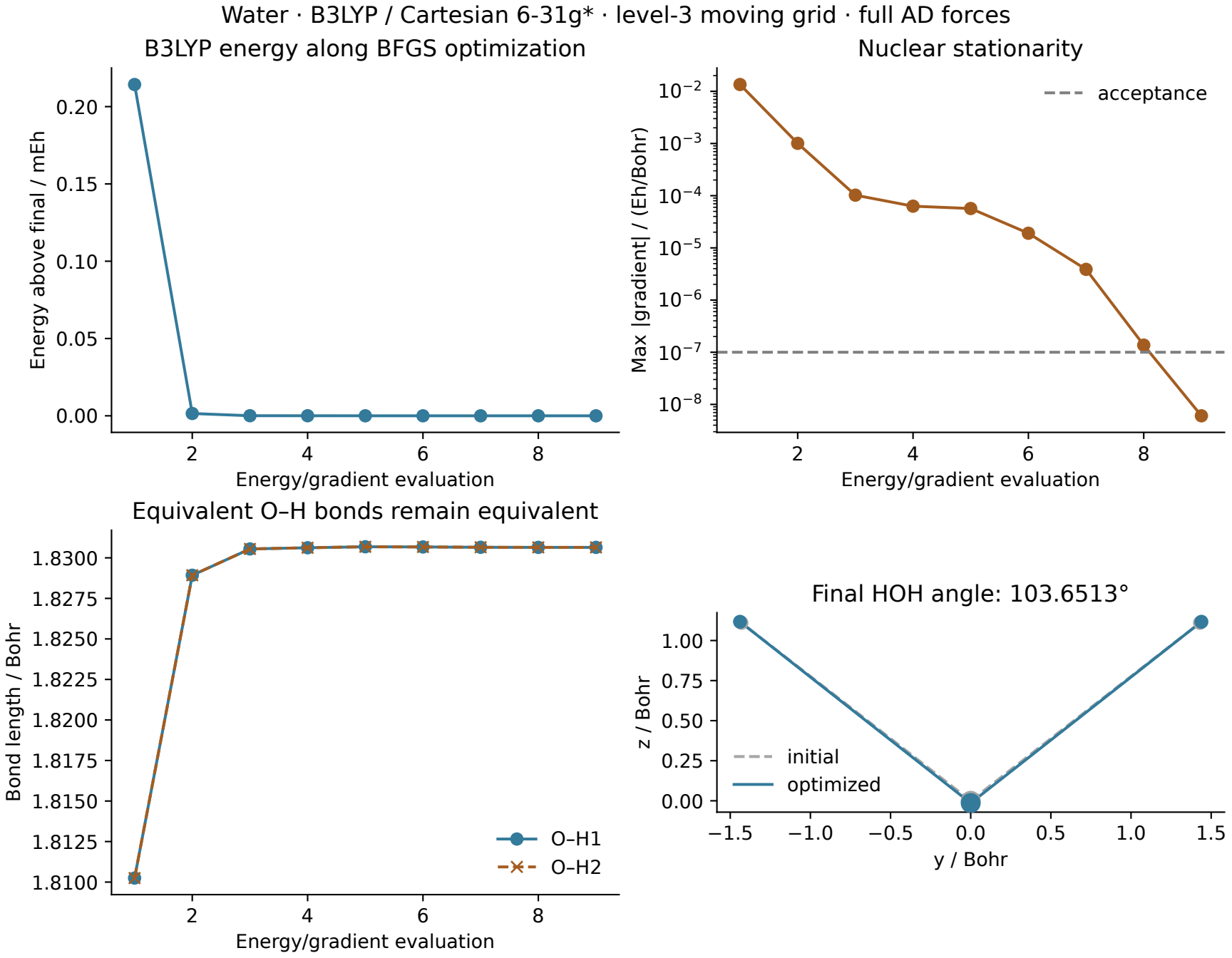
Water · B3LYP / Cartesian 6-31g* · level-3 moving grid · full AD forces
B3LYP energy along BFGS optimization
Energy above final / mEh
0.20
0.15
0.10
0.05
0.00
2
4
6
8
Energy/gradient evaluation
Nuclear stationarity
Max |gradient| / (Eh/Bohr)
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-7}
10^{-8}
acceptance
2
4
6
8
Energy/gradient evaluation
Equivalent O–H bonds remain equivalent
Bond length / Bohr
1.8300
1.8275
1.8250
1.8225
1.8200
1.8175
1.8150
1.8125
1.8100
O–H1
O–H2
2
4
6
8
Energy/gradient evaluation
Final HOH angle: 103.6513°
z / Bohr
1.00
0.75
0.50
0.25
0.00
initial
optimized
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
y / Bohr